Unit: 6 Unit 7: Factoring
Topic: Difference of Squares

Name: ________________
Date: ___________

Homework (Habanero)

Q1. Factor $x^2 - 4$
(A) $x^2 - 2^2$
(B) $(x-2)(x+2)$
(C) $x(x-4)$
(D) $x^2 - 2x + 4$
(E) $2(x^2 - 2)$

Q2. The area of a cell culture is given by $A = 9t^2 - 4t^2$. Factor the expression for A
(A) $5t^2$
(B) $5t^4$
(C) $5t^2(1+t^2)$
(D) $(3t-2t)(3t+2t)$
(E) $5t^2(1-t^2)$

Q3. The difference of squares of the lengths of the sides of a microscopic image are given by $64x^2 - 36x^2$. Factor the expression
(A) $(8x-6x)(8x+6x)$
(B) $28x^2$
(C) $28x^4$
(D) $2x^2(32-18)$
(E) $2x(8x-6x)(8x+6x)$

Q4. The amount of bacteria present in a hospital ward after applying a new sanitation method is modeled by $B = 25t^2 - 9t^2$. Factor the expression for B
(A) $16\mathrm{t}^4$
(A) $t(25-9t)$
(B) $4(25t^2 - 9t^2)$
(C) $16\mathrm{t}^2$
(C) $t^2(25-9)$

Q5. The square of the speed of a pharmaceutical delivery truck is given by $s^2 - 36$. Factor the expression
(A) $3s(s-4)$
(B) $(s-6)(s+6)$
(C) $36(\mathrm{s}^2 - 1)$
(D) $s(s-6)$
(E) $s(s+6)$

Q6. The expression $121x^2 - 25x^2$ represents the difference of squares. Factor the expression
(A) $x(11x-5x)(11x+5x)$
(B) $x^2(121-25)$
(C) $96x^2$
(D) $x(121-25)$
(E) $96x^2(1+x)$

Q7. The difference of squares of the growth rate of cancer cells are given by $r^2 - 9r^2$. Factor the expression
(A) $-8r^2$
(B) $r^2(r-3)$
(C) $8r(r+1)$
(D) $8r^2(r-1)$
(E) $-8r^2(r+3r)$

Q8. The area of a medical testing lab is given by $49m^2 - 16m^2$. Factor the expression for the area
(A) $33\mathrm{m}^2$
(A) $65\mathrm{m}^2$
(A) $m^2(49-16)$
(B) $33m(m-4)$
(C) $(7m-4m)(7m+4m)$

Open Ended Questions (FRQ)

Q9. The population growth of a new species of bacteria can be modeled by the equation $P = 16t^2 - 9t^2$. Factor the expression and explain its significance in the context of population growth.

Q10. A hospital is planning to expand its emergency ward. The area of the new ward is given by $A = 25x^2 - 16x^2$, where x is the length of the side of the ward. Factor the expression for A and calculate the area if $x = 4$ meters.

Chef's Tips

- Provide clear instructions
- Encourage critical thinking
- Remind students to check units of measurement